

Section 15.1: Double Integrals over Rectangles

Multiple Integrals

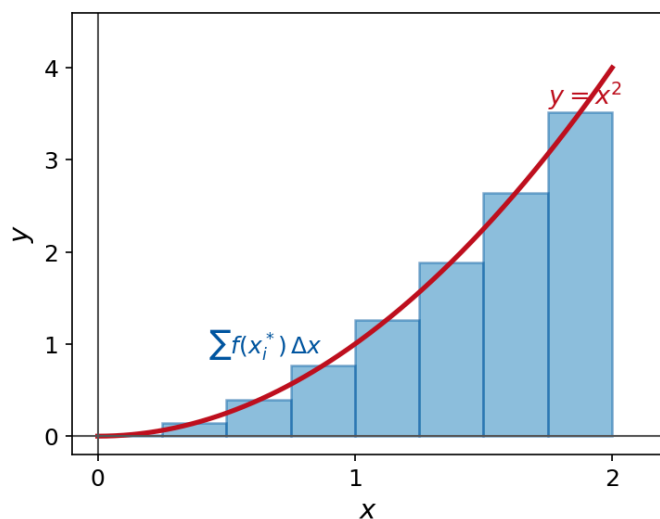
MTH 310
Calculus III

From Calc I to Calc III

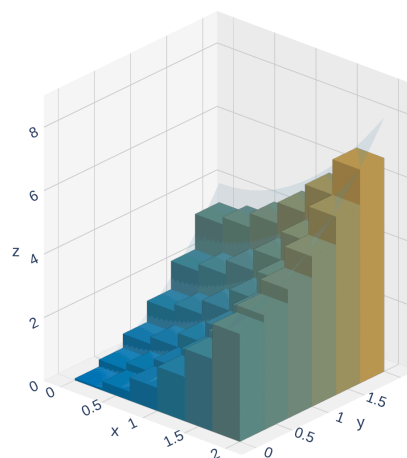
In Calculus I, the **area** under a curve $y = f(x)$ over $[a, b]$ is the limit of Riemann sums:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

Calc I: strips in 2D



Calc III: columns in 3D



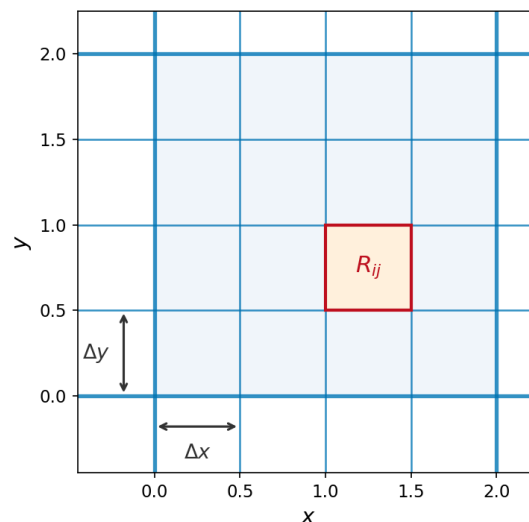
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Building a Double Riemann Sum

Suppose $f(x, y) \geq 0$ is defined on the rectangle $R = [a, b] \times [c, d]$.

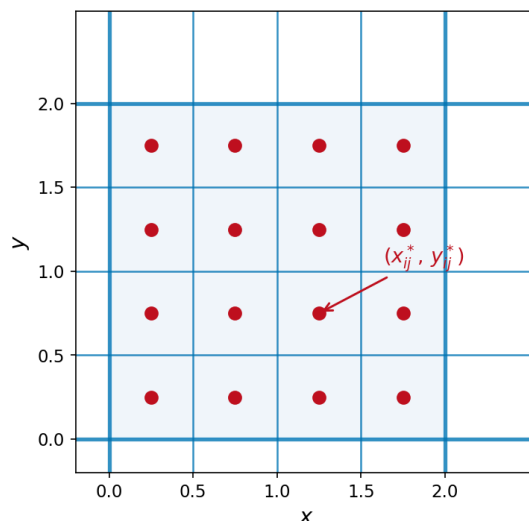
Step 1: Partition the rectangle.

- Divide $[a, b]$ into m equal subintervals of width $\Delta x = \frac{b-a}{m}$
- Divide $[c, d]$ into n equal subintervals of width $\Delta y = \frac{d-c}{n}$
- This creates $m \times n$ subrectangles, each with area $\Delta A = \Delta x \cdot \Delta y$



Step 2: Choose sample points.

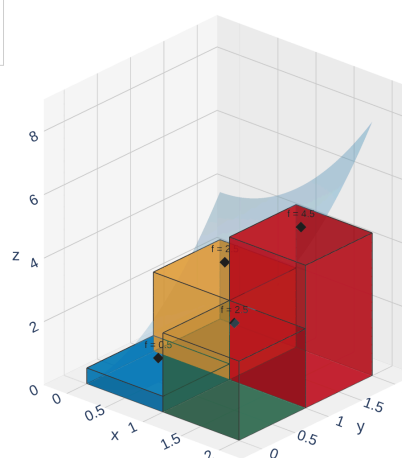
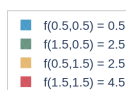
In each subrectangle R_{ij} , choose a sample point (x_{ij}^*, y_{ij}^*) . Common choices: corners, midpoints, or random points.



Step 3: Build columns and add.

Each column has base area ΔA and height $f(x_{ij}^*, y_{ij}^*)$, so its volume is $f(x_{ij}^*, y_{ij}^*) \Delta A$. The total volume is approximately:

$$V \approx \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$$



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The Double Integral

Double Integral over a Rectangle

The **double integral** of f over the rectangle R is:

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$$

provided this limit exists (in which case we say f is **integrable** over R).

Geometric interpretation:

- When $f(x, y) \geq 0$: $\iint_R f dA = \mathbf{volume}$ under the surface and above R
- When f takes both signs: $\iint_R f dA = (\text{volume above } R) - (\text{volume below } R)$

The double integral generalizes the single integral: $\int f(x) dx$ gives area under a curve; $\iint f(x, y) dA$ gives volume under a surface.

Example 1: Midpoint Approximation

Example 1

Let $f(x, y) = x^2 + y^2$ and $R = [0, 2] \times [0, 2]$. Approximate the volume under f over R by dividing R into a 2×2 grid and using midpoints.

Draw the 2×2 grid on $[0, 2] \times [0, 2]$. Find ΔA . Mark the midpoints. Evaluate f at each.

Your Turn: Midpoint Riemann Sum

Example 2

Let $f(x, y) = xy$ and $R = [0, 1] \times [0, 1]$. Approximate the volume under f over R using a 2×2 grid and midpoints.

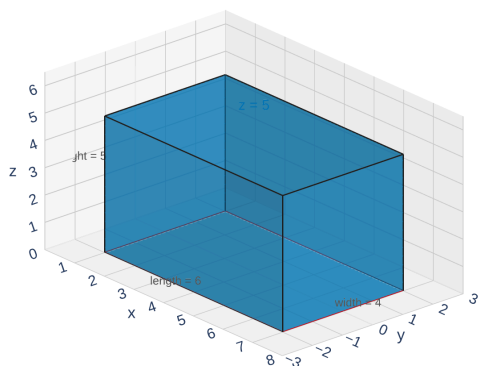
Draw the 2×2 grid on $[0, 1] \times [0, 1]$. Find ΔA . Mark the midpoints. Evaluate f at each.

Geometric Interpretation of the Double Integral

When $f(x, y) \geq 0$, the double integral $\iint_R f \, dA$ gives the **volume** of the solid between the surface $z = f(x, y)$ and the rectangle R in the xy -plane.

This means we can evaluate some double integrals by recognizing the solid as a **familiar geometric shape**.

■ $z = 5$
— $R = [1, 7] \times [-2, 2]$

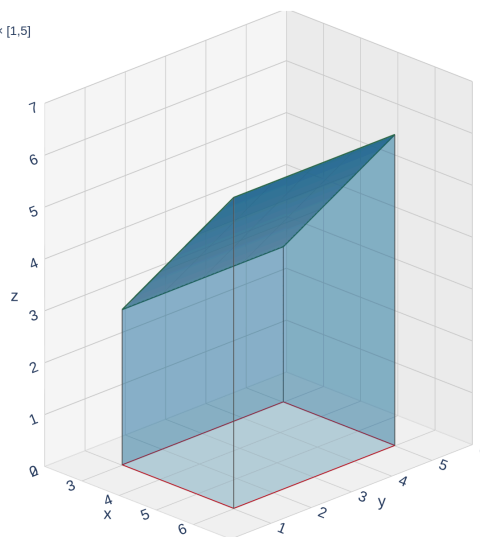


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Constant function: $z = c$ over R is a **box**.

Volume = length $\times$ width $\times$ height

— $R = [3, 6] \times [1, 5]$



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Linear function: $z = x$ over R is a **ramp/wedge**.

Volume = cross-sectional area $\times$ depth

Before reaching for computation, ask: **can I see the shape?** If the solid is a box, wedge, cylinder, cone, or hemisphere, use the geometric formula!

Example 2: Volume of a Box

Example 2

Evaluate $\iint_R 5 \, dA$ where $R = \{(x, y) \mid 1 \leq x \leq 7, -2 \leq y \leq 2\}$.

The surface $z = 5$ is a horizontal plane. What solid sits between this plane and the rectangle R ?

When the integrand is a **constant** c , $\iint_R c \, dA = c \cdot \text{Area}(R)$

Your Turn: Geometric Interpretation

Without computing any integrals, what is $\iint_R (1 - x) \, dA$ where $R = [0, 1] \times [0, 1]$?

Hint: Sketch $z = 1 - x$ over the unit square. What shape do you see?

The surface $z = 1 - x$ is a plane. At $x = 0$ the height is 1; at $x = 1$ the height is 0. What solid is this?

A linear function over a rectangle always produces a shape you can compute geometrically — no integral formulas needed!

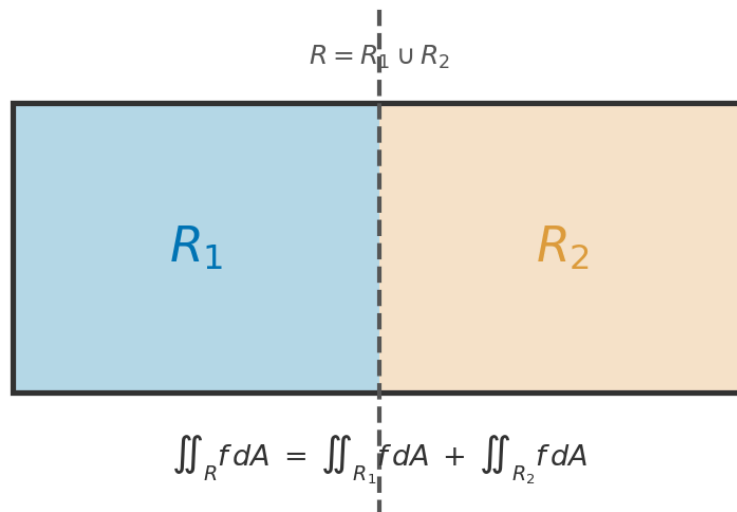
Properties of Double Integrals

Double integrals obey the same **linearity rules** as single integrals.

Theorem: Properties of Double Integrals

If f and g are integrable over R and c is a constant:

1. **Constant multiple:** $\iint_R cf \, dA = c \iint_R f \, dA$
2. **Sum/Difference:** $\iint_R (f \pm g) \, dA = \iint_R f \, dA \pm \iint_R g \, dA$
3. **Comparison:** If $f(x, y) \geq g(x, y)$ for all (x, y) in R , then $\iint_R f \, dA \geq \iint_R g \, dA$
4. **Splitting:** If $R = R_1 \cup R_2$ where R_1 and R_2 don't overlap (except on edges), then
$$\iint_R f \, dA = \iint_{R_1} f \, dA + \iint_{R_2} f \, dA$$



These properties follow directly from the corresponding properties of finite sums and limits. They'll be essential when we learn to **compute** double integrals via iterated integrals in the next section.

Linearity is your best friend: it lets you break complicated integrals into simpler pieces.

Looking Ahead: How Will We Actually Compute These?

So far, we've only evaluated double integrals by:

- Riemann sum approximations (Example 1)
- Geometric formulas for simple shapes (Example 2)

The big question: What about $\iint_R e^{x+y} dA$ or $\iint_R \sin(x^2 y) dA$? We can't draw those as boxes or wedges!

Coming in Section 15.2: Fubini's Theorem tells us we can compute a double integral as two nested single integrals:

$$\iint_R f(x, y) dA = \int_a^b \left[\int_c^d f(x, y) dy \right] dx$$

This is called an **iterated integral**, and it reduces the problem to two Calc I integrals!

Today's takeaway: understand **what** a double integral is (limit of Riemann sums, volume under a surface). Next time: learn **how** to compute them efficiently.

Homework

Section 15.1: 1-11 odd, 17, 19, 21, 25, 29, 31, 33, 45, 55